LIVE DATA IN POWER BI

MySQL Workbench → Power BI Desktop

Step-by-step classroom practical

Database	LiveSalesDB
Table	SalesTable
Server	localhost:3306
Authentication	Database
Power BI mode	Import + Refresh

Goal: connect MySQL data to Power BI, build visuals, insert new SQL rows, and refresh the report to display the new data.

OVERVIEW What You Will Build

You already created a MySQL database and inserted values. The remaining practical is to connect that database to Power BI, load SalesTable, create visuals, add a new row in MySQL, and refresh Power BI to show the new row.

Step	Action
1	Verify database and table
2	Check connection details
3	Install MySQL Connector/NET if needed
4	Connect Power BI to MySQL
5	Authenticate with Database credentials
6	Load SalesTable
7	Create table and chart visuals
8	Insert a new row in MySQL
9	Refresh Power BI and verify the update

Final flow:

```
MySQL Workbench
      ↓
LiveSalesDB → SalesTable
      ↓
Power BI MySQL Connector
      ↓
Import Data → Build Visuals
      ↓
INSERT new row in MySQL
      ↓
Power BI Refresh
      ↓
New row appears in visuals
```

STEP 1 Verify the Database and Table

Run these commands in MySQL Workbench before opening Power BI.

Select the database you created:

```
USE LiveSalesDB;
```

Check that SalesTable exists and contains your data:

```
SELECT * FROM SalesTable;
```

Your table can be as simple as:

```
CREATE TABLE SalesTable (
    SalesID INT PRIMARY KEY,
    Product VARCHAR(50),
    Quantity INT
);
```

Expected structure:

Column	Meaning
SalesID	Integer; unique ID for each sale
Product	Product name
Quantity	Number of units sold

STEP 2 Insert the Initial Sales Data

If you have already inserted values, you can simply verify them. Otherwise use:

```
INSERT INTO SalesTable
VALUES
(1, 'Laptop', 2),
(2, 'Mouse', 5),
(3, 'Keyboard', 3);
```

Check the result:

```
SELECT * FROM SalesTable;
```

SalesID	Product | Quantity
1	Laptop | 2
2	Mouse | 5
3	Keyboard | 3

At this point the SQL side is ready. Do not close MySQL Workbench; you will return to it later to insert new rows.

STEP 3 Check the MySQL Connection Details

Power BI needs four things: server/host, port, database name, and MySQL username.

Check the host name:

```
SELECT @@hostname;
```

On your computer this returned:

```
DESKTOP-J3PH8NS
```

Check the port:

```
SHOW VARIABLES LIKE 'port';
```

The default MySQL port is normally 3306.

Check the current database and user:

```
SELECT DATABASE();
SELECT USER();
```

Setting	Value
Server for this PC	localhost:3306
Alternative server	DESKTOP-J3PH8NS:3306
Database	LiveSalesDB
Authentication	Database
Username	Use the MySQL user returned by SELECT USER();, often root

STEP 4 Install the MySQL Connector if Power BI Requests It

Power BI Desktop may require Oracle MySQL Connector/NET to communicate with MySQL.

- If Power BI already opens the MySQL login screen, continue to Step 5.
- If Power BI says the MySQL provider/connector is missing, install MySQL Connector/NET.
- After installation, completely close Power BI Desktop and reopen it.

Important: you are connecting to MySQL, so in Power BI choose MySQL database - not SQL Server.

```
Correct:   Get Data → MySQL database
Wrong:     Get Data → SQL Server
```

STEP 5 Open the MySQL Connector in Power BI

Open Power BI Desktop, then follow:

```
Home
  ↓
Get data
  ↓
More...
  ↓
Search: MySQL
  ↓
MySQL database
  ↓
Connect
```

If you accidentally choose SQL Server, Power BI may show an error such as "Could not open a connection to SQL Server". That happens because MySQL and Microsoft SQL Server are different database systems.

STEP 6 Enter Server and Database

In the MySQL database connection window, enter:

Field	Enter
Server	localhost:3306
Database	LiveSalesDB

Because Power BI and MySQL are running on the same computer, localhost is the simplest server value.

You can also use your host name:

```
DESKTOP-J3PH8NS:3306
```

Recommended connection for your setup:

```
Server:    localhost:3306
Database:  LiveSalesDB
```

Click OK / Connect to continue.

STEP 7 Choose Database Authentication

This is the most important authentication step for your current setup.

Power BI may show two authentication choices on the left:

```
Windows
Database
```

Select Database. Do not use Windows credentials for this MySQL connection.

Enter the same MySQL username and password that you use in MySQL Workbench:

Field	Value
Authentication	Database
User name	Example: root
Password	Your MySQL password
Apply settings to	localhost:3306

To verify the MySQL username in Workbench, run:

```
SELECT USER();
```

Example output:

```
root@localhost
```

In that case, enter root as the Power BI username. Then click Connect.

STEP 8 Select SalesTable and Load It

After successful authentication, Power BI opens the Navigator window.

Expand your database and select:

```
LiveSalesDB
   └── SalesTable
```

Tick SalesTable. A preview should show your data, for example:

SalesID	Product | Quantity
1	Laptop | 2
2	Mouse | 5
3	Keyboard | 3

Click Load. Power BI imports the current data from MySQL into the report model.

STEP 9 Create the Power BI Visuals

First create a simple table visual. Add these fields:

```
SalesID
Product
Quantity
```

Then create a Clustered column chart:

Visual field	Use
X-axis	Product
Y-axis	Quantity

The report should now represent the same data you saw in MySQL Workbench.

```
Laptop      ▟▙      2
Mouse       ▟▙▙▙    5
Keyboard    ▟▙▙     3
```

Keep Power BI open. Next you will change the source database.

STEP 10 Insert a New Row in MySQL

Return to MySQL Workbench. Add a new sale:

```
INSERT INTO SalesTable
VALUES (4, 'Monitor', 6);
```

Verify the new row:

```
SELECT * FROM SalesTable;
```

MySQL should now contain:

SalesID	Product | Quantity
1	Laptop | 2
2	Mouse | 5
3	Keyboard | 3
4	Monitor | 6

The database has changed. Your Power BI report still contains the previously imported data until it is refreshed.

STEP 11 Refresh Power BI and See the New Data

Return to Power BI Desktop and select:

```
Home → Refresh
```

Power BI runs the query again, imports the latest data, and updates the visuals.

You should now see Monitor with quantity 6 in both the table and chart.

```
MySQL INSERT
      ↓
Database changes
      ↓
Power BI Refresh
      ↓
Latest rows loaded
      ↓
Visuals update
```

You can repeat the demonstration with another row:

```
INSERT INTO SalesTable
VALUES (5, 'Headphone', 10);
```

Then click Refresh in Power BI again.

IMPORTANT What Is "Live" in This MySQL Setup?

Your current MySQL → Power BI connection is an Import-based workflow. That means Power BI stores a copy of the data in its model.

Event	What happens
MySQL row changes	The source database changes immediately.
Power BI report	Still shows the last imported copy.
Power BI Refresh	Loads the latest MySQL rows and updates visuals.

So the teaching flow is:

```
INSERT new data in MySQL
        ↓
Click Refresh in Power BI
        ↓
Updated data appears
```

This is not the same as DirectQuery + Automatic Page Refresh. If you need a dashboard that checks the database every few seconds without pressing Refresh, use a source that supports DirectQuery for that scenario, such as Microsoft SQL Server.

TROUBLESHOOTING Common Problems and Fixes

Problem	Fix
"Could not connect to SQL Server"	You selected SQL Server. Go back and choose MySQL database.
Authentication failed	Choose Database authentication, not Windows. Use your MySQL username/password.
Not sure about username	Run SELECT USER(); in MySQL Workbench.
Not sure about database	Run SELECT DATABASE(); or SHOW DATABASES;.
Not sure about port	Run SHOW VARIABLES LIKE 'port';
Provider/connector missing	Install MySQL Connector/NET, then restart Power BI Desktop.
Table not visible	Run USE LiveSalesDB; and SHOW TABLES; in Workbench.
New row not appearing	Click Home → Refresh in Power BI.
localhost fails	Try DESKTOP-J3PH8NS:3306 and verify the MySQL service is running.

CLASSROOM RECAP Complete Practical in One Page

```
-- MySQL Workbench
USE LiveSalesDB;

SELECT * FROM SalesTable;

SELECT @@hostname;
SHOW VARIABLES LIKE 'port';
SELECT USER();

INSERT INTO SalesTable
VALUES (4, 'Monitor', 6);

SELECT * FROM SalesTable;
```

Power BI connection:

```
Get Data → MySQL database
Server: localhost:3306
Database: LiveSalesDB
Authentication: Database
Username: your MySQL username
Password: your MySQL password
Select SalesTable → Load
```

Final demonstration:

```
1. Show existing Power BI visuals
2. Insert a new row in MySQL
3. Show SELECT * FROM SalesTable
4. Return to Power BI
5. Click Home → Refresh
6. Show the new row in the table/chart
```